

Open Questions of the Emergent Gravity Program: Attempts, Partial Results, and Admissions

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Abstract

The Emergent Gravity Program’s Final Status table of 14 August 2026 closes the linearized Einstein–Cartan certification inside holographic AdS and leaves four items open: the multi-digit coefficient of the kinematic residue I_B , the de Sitter / cosmological domain, nonlinear Einstein–Cartan from entanglement, and UV completion of the effective Lagrangian. This note attempts each item in turn.

One computation is performed: a $d = 2$ hard-cutoff subtraction of the coincident $1/(x-y)^2$ piece of the flowed chiral-current two-point function against the modular kernel $\pi/(2 \cosh^2(\pi s))$. Three grids give a relative residue $r \in \{-0.656, -0.622, -0.645\}$, spread 0.034. That is consistent with the order-unity claim already on the books. It is *not* a multi-digit coefficient. Paper VIII already stated that further numerical proxies of this class will not produce one. This note confirms that statement by failing in the same way.

The other three items are not computed. A literature audit shows: (i) no Casini–Huerta–Myers modular Hamiltonian of equal rigor exists for generic de Sitter regions; (ii) the Faulkner–Haehl–Hijano–Parrikar–Rabideau–Van Raamsdonk second-order derivation [5] yields Einstein’s equations, not Einstein–Cartan, and Jacobson’s entanglement-equilibrium proposal [6] remains a conjecture for the full nonlinear theory; (iii) UV completion is open by construction of the effective Lagrangian. Each failure is recorded as an **[ADMISSION]**, not papered over.

This note does not move any Final Status label from **[O]** to **[P]**.

How to read this note

Every material sentence is one of: a restatement of a labeled claim already in the 14 August series; a computation whose script and number are cited; a citation to a published paper with a location; or an **[ADMISSION]**. No sentence in §§3–6 upgrades an **[O]** to a **[P]**.

This draft was prepared with an AI research agent under the program’s standing rule that model output is a hypothesis until verified by a derivation, a script, or the author. The I_B numbers below come from a script. The de Sitter, nonlinear, and UV sections are literature audits, not derivations. The author owns any claim he endorses.

1 The four opens, quoted

The Final Status table [21] records, verbatim in substance:

Item	Label
Multi-digit coefficient of I_B	[O]
de Sitter / cosmological domain	[O]
Nonlinear regime (no derivation at FGHMV standard)	[O]
UV completion (open by construction)	[O]

Paper VIII [19] adds the operational criterion for the first item: a fully analytic Hadamard (or equivalent distributional) expansion of the current two-point function under modular flow, folded against

$$K(s) = \frac{\pi}{2 \cosh^2(\pi s)}, \quad \int_{-\infty}^{\infty} K(s) ds = 1. \quad (1)$$

Paper IX [20] adds the operational criterion for the second: a family of de Sitter regions whose modular Hamiltonians are under analytic control, a first-law identity, equivalence to linearized Einstein–Cartan including the algebraic torsion equation, and recovery of the Hehl–Datta term from relative-entropy positivity. The third and fourth items have no such criterion in the series beyond “a derivation at FGHMV standard” and “not claimed.”

2 What is not reopened

For the avoidance of doubt, this note does *not* re-litigate:

- FGHMV: ball first law $\Leftrightarrow$ linearized Einstein about AdS [1]. [P] as external literature.
- The modular perturbation kernel of Paper IV and its machine-check. [P] in the series; not recomputed here.
- Unconditional metric sector of Condition NL (conformal selection rule). [P] in the series.
- $I_B \not\equiv 0$ for finite regions, relative size $O(1)$, absorbable. [P] in Papers VI and VIII.
- Hehl–Datta sign from relative-entropy positivity; magnitude $-3\kappa/16$ by rigidity and by the null-plane + absorbability argument of Paper VII. [P] in the series.
- The algebraic Einstein–Cartan elimination $\mathcal{L}_{\text{HD}} = -3\kappa/16 J_5 \cdot J_5$ as a theorem of Palatini + Dirac, independently extracted in-platform at residual $< 10^{-15}$ (OpenWave M9.1). That extraction does not select Einstein–Cartan from entanglement; it only recovers the 1971 identity.

3 Attempt I: the multi-digit coefficient of I_B

3.1 Definition

Paper VI writes the surviving axial channel as $N_B[\beta; \zeta] = C_J I_B[\beta; \zeta]$ with

$$I_B[\beta; \zeta] = - \int_{-\infty}^{\infty} ds K(s) \langle \sigma_s J_5[\beta] \tilde{J}_5[\zeta] \rangle_{\kappa \neq 0}. \quad (2)$$

In $d = 2$ chiral CFT the interval modular flow is the Möbius action [17]

$$x(s) = R \tanh(\operatorname{artanh}(x/R) + \pi s), \quad (3)$$

the current two-point function is

$$\langle J(x)J(y) \rangle = \frac{k}{(x-y)^2} \quad (4)$$

for a normalization k that cancels in any relative residue, and the weight-one Jacobian is $J(s, x) = \partial x(s)/\partial s$.

The local quadratic form is the $s = 0$ value of the smeared correlator. The relative residue reported below is

$$r := \frac{\int ds K(s)(Q(s) - Q(0))}{|Q(0)|}, \quad Q(s) = \int dx dy \beta(x)\beta(y) \frac{J(s, x)}{(x(s) - y)^2}. \quad (5)$$

A compactly supported bump $\beta(x) = \exp(-1/(1 - (x/a)^2))$ for $|x| < a < R$ is used.

3.2 What a Hadamard extraction would be

Write the flowed two-point function as a distribution on $x, y \in (-R, R)$. A Hadamard expansion isolates

$$\langle \sigma_s J(x) J(y) \rangle = \frac{u_2(s; x, y)}{(x - y)^2} + \frac{u_1(s; x, y)}{x - y} + v(s; x, y) \log |x - y| + w_{\text{fin}}(s; x, y), \quad (6)$$

with $u_2, u_1, v, w_{\text{fin}}$ smooth, then subtracts u_2, u_1, v by a finite-part prescription that commutes with the s -integral against $K(s)$. The number of record would be the integral of $K(s)w_{\text{fin}}$ (smeared), independent of the cutoff to the quoted digits.

3.3 What was actually computed

A hard spatial cutoff ε on $|x(s) - y|$ was imposed and r was evaluated on three grids (`m9_2_ib_hadamard.py`):

$(N_x, N_s, s_{\text{max}}, \varepsilon)$	r
(201, 61, 2.0, 0.04)	−0.6555
(241, 81, 2.5, 0.03)	−0.6216
(281, 81, 2.5, 0.02)	−0.6450

Spread: 0.034. Kernel integral on these grids is 1 to $O(10^{-3})$ (truncation in s). The three values lie in the order-unity window already reported by Paper VIII (“approximately 1–5 depending on the finite-part implementation”). The sign here is an artefact of reporting r as $\int K(Q - Q_0)/|Q_0|$ with this Jacobian convention; the structural statement is $|r| = O(1)$, not a signed high-precision coefficient.

3.4 Why this is not a Hadamard extraction

Admission. The u_1 and log terms of (6) were not isolated. The s -dependence of u_2 was not subtracted. The cutoff ε remains visible at the third digit. No three-digit agreement exists across the three grids. Paper VIII already closed the numerical campaign for this reason. Repeating a discarded proxy does not reopen it. The multi-digit coefficient of I_B remains **[O]**.

A mutation that replaced $K(s)$ by a flat measure on $[-s_{\text{max}}, s_{\text{max}}]$ would change r by an $O(1)$ factor; that check was not needed to establish failure, because the scheme spread already exceeds any honest claim of “digits.”

3.5 What would close Q1

An analytic computation of w_{fin} for the Möbius flow of $\langle J(x)J(y) \rangle$, or a numerical scheme whose finite part is stable to ≥ 3 digits under simultaneous refinement of (N_x, N_s, ε) and under at least two inequivalent subtractions (Hadamard vs. differential regularization). Until one of those exists, no label change is licensed.

4 Attempt II: the de Sitter domain

4.1 What AdS supplies and de Sitter does not

Paper IX lists four ingredients used by every [P] step of the series [20]:

1. the Casini–Huerta–Myers modular Hamiltonian for balls, which is a conformal image of the Rindler boost [2];
2. a global conformal structure on the boundary;
3. a clean asymptotic boundary with a holographic dictionary [10];
4. a Ryu–Takayanagi surface whose modular flow is a bulk Killing horizon [8, 9].

de Sitter has a cosmological horizon and a Gibbons–Hawking entropy [12]. It does not have (1)–(4) in the same form. Static-patch modular Hamiltonians exist as operator statements about the static observer algebra; they are not given by a CHM-type local integral of T_{00} on an arbitrary ball. There is no generally accepted microscopic dual that would supply an FGHMV equivalence [11]. The classical static-patch first law carries a characteristic minus sign: adding Killing energy *decreases* horizon entropy [13]. That sign is already a structural obstruction to copying the AdS bookkeeping line by line.

4.2 What was attempted

No de Sitter modular Hamiltonian was derived in this note. No static-patch first-law $\Leftrightarrow$ linearized Einstein–Cartan map was written. No Hehl–Datta coefficient was recovered from relative entropy on a cosmological horizon.

A literature pass confirms that recent work on de Sitter entanglement (timelike extremal surfaces, c -theorems in a fixed de Sitter background, static-patch algebra) does not supply the four items of Paper IX. Citing those papers as if they closed Q2 would be a category error.

Admission. Q2 is unattempted as a derivation and unclosed as a theorem. The AdS results stand. Their extrapolation to cosmology remains [O]. This note adds no new de Sitter identity.

4.3 What would close Q2

Exactly the four-point checklist of Paper IX [20]: controlled modular Hamiltonians on a family of de Sitter regions; a first-law identity; equivalence to linearized Einstein–Cartan including $T^\lambda_{\mu\nu} + \dots = \kappa s^\lambda_{\mu\nu}$; and the Hehl–Datta term from relative-entropy positivity. Anything weaker is a hope.

5 Attempt III: nonlinear Einstein–Cartan

5.1 What exists for Einstein gravity

At first order, FGHMV [1] is the theorem the program uses. At second order, Faulkner, Haehl, Hijano, Parrikar, Rabideau and Van Raamsdonk [5] show that if ball entanglement entropies of a restricted class of CFT states are represented by the HRRT formula in an asymptotically AdS geometry, that geometry satisfies Einstein’s equations through second order, with matter determined by the CFT one-point functions, under the central-charge relation $\tilde{C}_T = a_*$. The derivation uses the Hollands–Wald identity [7] and the quantum Fisher / canonical-energy matching of Lashkari–Van Raamsdonk [4].

That is a real theorem about *Einstein* gravity. It is not a theorem about Einstein–Cartan gravity. The HRRT functional is an area. The Hollands–Wald charge used in [5] is the Einstein–Hilbert Noether charge. Torsion does not appear.

Jacobson’s entanglement-equilibrium argument [6] proposes that the full nonlinear Einstein equation is the equilibrium condition of entanglement on small balls, for a maximally symmetric vacuum. The series already tags the transfer of that argument to general backgrounds and to torsion as [C]. Thermodynamic derivations with torsion exist at the Clausius level [14]. They are not FGHMV-grade equivalences.

5.2 What was attempted

No second-order Einstein–Cartan identity was derived. No Hollands–Wald charge for the Palatini–Holst or Einstein–Cartan Lagrangian was written. No ball-by-ball matching of Fisher information to a torsional canonical energy at $O(\lambda^2)$ was performed.

Admission. Q3 remains [O]. The honest strongest statement this note can make is: nonlinear Einstein gravity from entanglement is a published theorem through second order [5]; nonlinear Einstein–Cartan from entanglement is not. Promoting the former to the latter by “the same bookkeeping” is a conjecture, and the series already refused to call it a theorem.

5.3 What would close Q3

A derivation, at the standard of [5] or of FGHMV, that the finite-ball entanglement first law (or relative entropy through second order) is equivalent to the nonlinear Einstein–Cartan equations, including the algebraic Cartan equation for torsion, with the Hehl–Datta contact generated rather than inserted. Until that exists, the nonlinear regime stays [O].

6 Attempt IV: UV completion

The New Standard Model is asserted to be emergent bookkeeping for a deeper description [16]. The deeper description, in the certified regime, is a holographic pair. No first-principles selection of that pair exists. Gauge group, representations, Yukawas, generations, and Λ are empirical input.

Admission. Q4 is open by construction. This note does not propose a CFT, a string compactification, an asymptotic-safety fixed point, or any other UV completion. Inventing one to fill the section would be a fabrication. The label remains [O].

A useful, and weaker, statement that *is* available: the effective Lagrangian (7) below is an EFT whose cutoff is at most M_{Pl} , whose new interaction is Planck-suppressed, and whose gravitational sector is certified only at linear order in AdS. That is a domain statement, not a completion.

$$S_{\text{NSM}} = \frac{1}{4\kappa} \int \varepsilon_{abcd} e^a \wedge e^b \wedge R^{cd} - \frac{\Lambda}{6} \int \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d + \int e \mathcal{L}_{\text{SM}}. \quad (7)$$

7 Cross-check: what this note must not be used for

- Do not cite §3 as a multi-digit value of I_B .
- Do not cite §4 as a de Sitter derivation.
- Do not cite §5 as nonlinear Einstein–Cartan.
- Do not cite §6 as a UV completion.
- Do not move any cell of an OpenWave MODELS.md gravity row on the strength of this note. The only in-platform number this program currently owns is the algebraic Hehl–Datta coefficient (M9.1), which is not one of the four opens.

8 Admissions, collected

Open	This note	Why it failed
Multi-digit I_B	attempted; $r \sim -0.63$, spread 0.034	hard cutoff is not Hadamard; digits move
de Sitter domain	not derived	CHM / RT / dictionary ingredients ab- sent
Nonlinear EC	not derived	published second-order theorem is Ein- stein, not EC
UV completion	not proposed	open by construction; no pair selected

No Final Status label is changed.

9 What remains worth doing

The only open that is a computation rather than a missing framework is Q1. The next honest attempt is analytic: expand $\langle J(x(s))J(y) \rangle$ for the flow (3) in the Hadamard form (6) and integrate $K(s)w_{\text{fin}}$ against a Schwartz source. If that integral can be reduced to a standard special function, the coefficient exists as a number. If the finite part depends on a renormalization scheme that the modular kernel does not fix, the coefficient is not universal and that fact should be recorded as a result, not hidden.

Q2–Q4 require ideas that this note does not contain.

Acknowledgments

Computational assistance was used to draft this note, to implement `m9_2_ib_hadamard.py`, and to check citations against the public record. All epistemic labels, including every **[ADMISSION]**, are the responsibility of the author. Numerical agreement is not proof. Consensus is not proof.

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